

C-convex.



c-conjugate

optimize only over $\mathcal{C}$ -convex functions

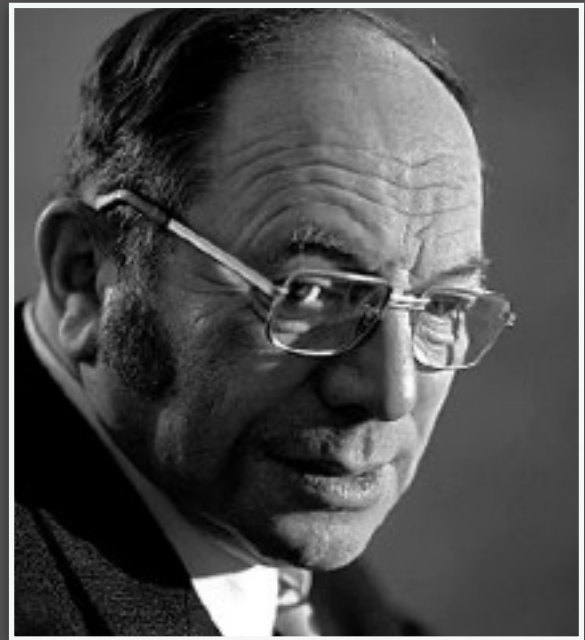
c -conjugacy

- By argument above, the optimal f, g must satisfy:

$$\forall x : f(x) = g^c(x) = \inf_y (c(x, y) - g(y)); \quad \forall y : g(y) = f^c(y) = \inf_x (c(x, y) - f(x));$$

- Not all functions can satisfy *both*. Those that do are called **c -convex**.
($\equiv$ that satisfy $(f^c)^c = f, (g^c)^c = g$)
- In that case, we say that (f, f^c) are **c -conjugate** (same for (g, g^c))
- Now, we can restrict problem to **optimize only over c -convex functions**, and replace g by f^c (or f by g^c).

A reformulated dual



Kantorovich's Problem:

(Continuous version dual)

$$\sup_{f,g : f+g \leq c} \int_X f \, d\alpha + \int_Y g \, d\beta$$